

Causal Offset Bounded TE Curve Verification Screen Closeout

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This closeout records the bounded TE Curve Verification Pipeline screen for the `causal_offset_mean_calibration_pilot_2026_07_22` forward candidates on `polished_dataset` setpoint inputs. The screen compared the two causal offset / mean calibration candidates against the accepted polished-setpoint forward baselines and the recent scalar shape-objective high-water reference.

Execution Summary

Item	Value
Screen	<code>causal_offset_bounded_track2_screen_2026_07_22</code>
Dataset	<code>polished_dataset</code>
Input mode	<code>setpoints</code>
Surface	<code>Fw / forward</code>
Candidate count	<code>5</code>
Curve count	<code>100</code>
Matrix output	<code>output/validation_checks/track2_reference_comparison/2026-07-23-13-07-44__causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_causal_offset_bounded_track2_screen_polished_setpoints_fw</code>
Matrix report	<code>doc/reports/analysis/validation_checks/te_curve_verification_pipeline/2026-07-23-13-08-58_causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_causal_offset_bounded_track2_screen_polished_setpoints_fw_report.md</code>
Reranker output	<code>output/validation_checks/shape_gated_te_curve_reranker/2026-07-23-13-09-03__shape_gated_te_curve_reranker</code>
Reranker report	<code>doc/reports/analysis/te_curve_verification_pipeline/03_cvp_diagnostics/shape_gated_reranker/[2026-07-22]/causal_offset_bounded_track2_screen_polished_setpoints_fw_matrix_shape_gated_te_curve_reranker_report.md</code>
Plot count	<code>0</code>

Metric Ranking

Rank	Candidate	Raw MAE deg	RMSE deg	Mean error %	P95 error %
1	<code>polished_setpoints_</code>	0.001837	0.002179	3.750629	8.248098

Rank	Candidate	Raw MAE deg	RMSE deg	Mean error %	P95 error %
	periodic_gru_ sequence_Fw				
2	polished_ setpoints_ periodic_mlp_ harmonic_Fw	0.001938	0.002276	3.987765	10.017324
3	shape_ objective_ periodic_mlp_ harmonic_Fw	0.002035	0.002412	4.222146	10.956483
4	causal_offset_ mean_periodic_ mlp_harmonic_Fw	0.002075	0.002478	4.314268	9.717112
5	causal_offset_ mean_gru_ sequence_Fw	0.002392	0.002868	5.009841	10.269009

Shape-Gated Decision

Rank	Candidate	Centered MAE deg	Shape pass	Composite	Decision
1	polished_ setpoints_ periodic_gru_ sequence_Fw	0.001483	0.950000	0.007008	recommended
2	polished_ setpoints_ periodic_mlp_ harmonic_Fw	0.001490	0.920000	0.239545	candidate
3	shape_ objective_ periodic_mlp_ harmonic_Fw	0.001578	0.910000	0.419662	candidate
4	causal_offset_ mean_periodic_ mlp_harmonic_Fw	0.001652	0.920000	0.424363	candidate
5	causal_offset_ mean_gru_ sequence_Fw	0.002024	0.000000	0.961925	shape gate failed

Harmonic Breakdown

Rank	Candidate	FFT sim.	Amp err. %	Phase err. deg
1	polished_setpoints_ periodic_gru_ sequence_Fw	0.984971	17.554759	12.349713
2	polished_setpoints_ periodic_mlp_ harmonic_Fw	0.983739	15.623668	12.951608
3	shape_objective_ periodic_mlp_ harmonic_Fw	0.983709	18.390567	17.439607
4	causal_offset_mean_ periodic_mlp_ harmonic_Fw	0.983672	19.654545	18.516624

Rank	Candidate	FFT sim.	Amp err. %	Phase err. deg
5	causal_offset_mean_ gru_sequence_Fw	0.975176	77.578652	71.215158

Interpretation

The bounded screen does not promote either causal offset / mean calibration candidate.

The accepted `polished_setpoints_periodic_gru_sequence_Fw` baseline remains the forward recommendation. It leads on raw MAE, centered MAE, shape pass rate, and composite score.

The non-windowed causal offset MLP remains a valid completed pilot, but it ranks behind the accepted GRU, the non-windowed polished-setpoint MLP baseline, and the prior shape-objective MLP reference. It does not justify expansion from this profile.

The time-windowed causal residual-offset GRU is rejected for this branch. Its shape pass rate is `0.000000`, and the reranker reports failures dominated by harmonic amplitude error, phase error, derivative agreement, FFT similarity, and peak-to-peak error.

Output Quality Finding

The matrix summary reports `report_plot_count: 0` with empty preview and report plot lists. Therefore this screen has scalar and shape-gated evidence, but it does not have the expected Track 2 measured-versus-predicted plot package.

The local operator log `.temp/campaing_log.log` also contains very long unwrapped transfer/progress lines and PowerShell host records. The most visible source is interactive `scp` transfer progress emitted during remote sync. This is an operator-output problem, not a model-result change, and should be fixed permanently in the campaign launcher/tooling after this closeout commit.

Decision

Do not promote the causal offset / mean calibration candidates.

Do not expand the causal residual-offset GRU branch from this evidence.

Keep both time-windowed and non-windowed roads as comparison categories for future designs, but close this direct causal-offset mean-calibration profile as non-promoted.

The next repository step is to repair bounded campaign output ergonomics: remote sync logs should not flood the terminal with unwrapped progress lines, and bounded Track 2 screens should reliably produce the expected measured-versus predicted TE curve plots when plot generation is part of the operator-facing reporting contract.